Enterprise Network Security Design with VLAN Segmentation, ACL Enforcement, and Attacker Isolation – Cisco Packet Tracer Simulation

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Executive Summary

This project documentation outlines a small to medium-sized enterprise (SME) network simulation in Cisco Packet Tracer. The design shows department-based VLAN segmentation to isolate traffic between HR, Finance, IT, and Management departments, reducing risks like lateral movement and unauthorized access attempts. Critical servers (Web, DNS, and FTP) are placed in a dedicated VLAN (50) with restricted access. A separate Attacker/Test VLAN (99) is included to simulate threats such as ping sweeps and unauthorized access attempts.

Security is enforced through extended Access Control Lists (ACLs) applied inbound on Switch Virtual Interfaces (SVIs) for granular, least-privilege control:

- HR: Limited to Web (HTTP/HTTPS), FTP, and DNS access; no SSH access to the core switch
- Finance: Limited to Web (HTTP/HTTPS), FTP, and DNS access; no SSH access to the core switch.
- IT: Full access to all internal resources, including SSH to the core switch.
- Management: Full access to all internal resources, including SSH to the core switch.
- Attacker: Completely blocked from all internal VLANs.

Secure remote access via SSH is implemented exclusively on the core multilayer switch (CORE-SW), using SSH version 2, RSA keys, timeouts, retries, and a standard ACL (SSH-MGMT) that restricts access to only IT (192.168.30.0/24) and Management (192.168.40.0/24) subnets. Inter-VLAN routing is handled by the multilayer switch (Cisco 3560) via SVIs, with two Layer 2 access switches (Cisco 2960) for endpoint connectivity.

Objectives / Goals

- Implement VLAN-based segmentation for enterprise departments.
- Prevent lateral movement between HR and Finance networks.
- Restrict access to servers based on department role.
- Fully isolate the attacker VLAN from internal resources.
- Restrict access to each VLAN using extended ACLs.
- Demonstrate correct ACL placement and behavior.
- Validate security policies through functional testing.

Network Departments & Endpoints

The lab environment is segmented using VLAN-based network segmentation to simulate a real enterprise architecture.

VLAN 10 – HR Department

- HR-PC1
- HR-PC2
- HR-PC3
- HR-Laptop

  Total Endpoints: 4

VLAN 20 – Finance Department

- FIN-PC1
- FIN-PC2
- FIN-PC3
- FIN-Laptop

  Total Endpoints: 4

VLAN 30 – IT Department

- IT-PC1
- IT-PC2
- IT-PC3
- IT-Laptop

  Total Endpoints: 4

VLAN 40 – Management

- MGMT-PC
- MGMT-Laptop

  Total Endpoints: 2

VLAN 50 – Critical Servers Zone

- Web Server (HTTP/HTTPS enabled)
- DNS Server (A records configured)
- File Server (FTP enabled for testing)

  Total Servers: 3

VLAN 99 – Attacker/Test Network

- Attacker-PC

  Total Endpoints: 1

Total Endpoints in the Environment

- User endpoints: 14
- Servers: 3
- Attacker systems: 1

  Grand Total: 18 systems

Lab Assumptions

The following assumptions apply to this enterprise network simulation:

1. IP Addressing

- All devices are configured using **static IP addressing**.
- No DHCP services are deployed in this environment.
- Default gateways are manually assigned per VLAN.

2. Network Scope

- The lab simulates an **internal enterprise LAN environment only**.
- There is **no external WAN or Internet connectivity** configured.
- Security testing focuses strictly on **internal threats and lateral movement** scenarios.

3. Simulation Platform

- The environment is built using the latest available version of **Cisco Packet Tracer (2026 release)**.

- Some advanced real-world enterprise features may be limited due to simulator constraints.

4. Remote Access & Management

- SSH access is configured exclusively on the Core Multilayer Switch (CORE-SW**)**.

- SSH Version 2 with RSA key encryption is enforced.

- A standard ACL (SSH-MGMT) restricts access to:

    - IT subnet (192.168.30.0/24)

    - Management subnet (192.168.40.0/24)

- Servers do **not** have SSH enabled.

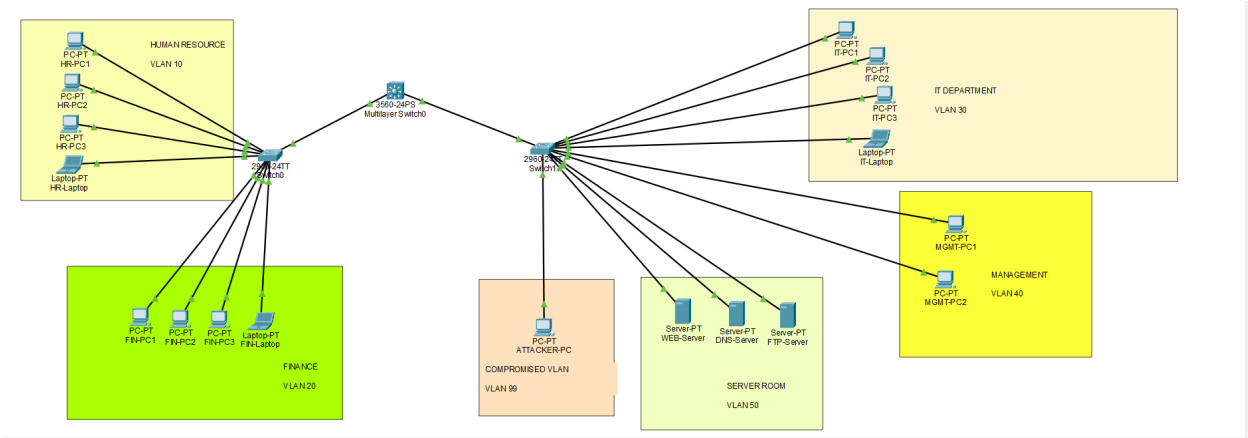
- No Telnet access is permitted anywhere in the network.

Network Design Overview

Department and Vlans

VLAN	Department	Subnet	Gateway	Device IP
10	HR	192.168.10.0/24	192.168.10.1	HR-PC1: 192.168.10.10 HR-PC2: 192.168.10.20 HR-PC3: 192.168.10.30 HR-Laptop: 192.168.10 .40
20	Finance	192.168.20.0/24	192.168.20.1	FIN-PC1: 192.168.20.10 FIN-PC2: 192.168.20.20 FIN-PC3: 192.168.20.30 FIN-Laptop:192.168.20.40
30	IT	192.168.30.0/24	192.168.30.1	IT-PC1: 192.168.30.10 IT-PC2: 192.168.30.20 IT-PC3:192.168.30.30 IT-Laptop:192.168.30.40
40	Management	192.168.40.0/24	192.168.40.1	MGMT-PC:192.168.40.10 MGMT-Laptop:192.168.40 .20

50	Servers	192.168.50.0/24	192.168.50.1	FTP Server: 192.168.50.10 DNS Server: 192.168.50.11 WEB Server: 192.168.50.12
99	Attacker	192.168.99.0/24	192.168.99.1	Attacker PC: 192.168.99.10



Description:

Core Multilayer Switch (Cisco 3560, hostname CORE-SW): Central device for Layer 3 routing.

- Connected to Switch1 via Gig0/1 (trunk, allowed VLANs: 10, 20, 30, 40, 50, 99).
- Connected to Switch2 via Gig0/2 (trunk, allowed VLANs: 10, 20, 30, 40, 50, 99).

Access Switch1 (Cisco 2960, hostname Access-SW1): HR & Finance.

- Fa0/1–4: HR devices (access VLAN 10).
- Fa0/5–8: Finance devices (access VLAN 20).
- Gig0/1: Trunk to CORE-SW.

Access Switch2 (Cisco 2960, hostname Access-SW2): IT, Management, Servers, Attacker.

- Fa0/1–4: IT devices (access VLAN 30).
- Fa0/5–6: Management devices (access VLAN 40).
- Fa0/7–9: Servers (access VLAN 50; Fa0/7 Web, Fa0/8 DNS, Fa0/9 File).
- Fa0/10: Attacker PC (access VLAN 99).
- Gig0/1: Trunk to CORE-SW.

Design & Configuration Details (Full CLI Commands)

1. Access Switch1 (Access-SW1) Configuration

```
hostname Access-SW1

vlan 10
 name HR
vlan 20
 name Finance

interface range fastEthernet 0/1 - 4
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown

interface range fastEthernet 0/5 - 8
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 no shutdown

interface gigabitEthernet 0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20
no shutdown

end
write memory
```

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/2
10   HR                               active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
20   FINANCE                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
```

3. Access Switch2 (Access-SW2) Configuration

```
hostname Access-SW2

vlan 30
 name IT
vlan 40
 name Management
vlan 50
 name Servers
vlan 99
 name Attacker

interface range fastEthernet 0/1 - 4
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 no shutdown

interface range fastEthernet 0/5 - 6
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
 no shutdown

interface range fastEthernet 0/7 - 9
 switchport mode access
 switchport access vlan 50
 spanning-tree portfast
 no shutdown
```

```
interface fastEthernet 0/10
 switchport mode access
 switchport access vlan 99
 spanning-tree portfast
 no shutdown

interface gigabitEthernet 0/1
 switchport mode trunk
 switchport trunk allowed vlan 30,40,50,99
 no shutdown

end
write memory
```

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
30   IT                               active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
40   MANAGEMENT                       active    Fa0/5, Fa0/6
50   SERVERS                          active    Fa0/7, Fa0/8, Fa0/9
99   ATTACKER                         active    Fa0/10
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

3. Core Multilayer Switch (CORE-SW) Configuration

Enter Configuration mode:

```
enable
configure terminal
```

Enable Layer 3 Routing

```
ip routing
```

Without this, **nothing routes**.

Create Vlans:

```
hostname CORE-SW
ip routing
vlan 10
 name HR
vlan 20
 name Finance
vlan 30
 name IT
vlan 40
 name Management
vlan 50
 name Servers
vlan 99
 name Attacker
```

```
CORE-SW>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
10   HR                               active
20   FINANCE                          active
30   IT                               active
40   MANAGEMENT                       active
50   SERVERS                          active
99   ATTACKER                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
CORE-SW>
```

VLANS

Create SVIs (Gateways):

HR VLAN 10

```
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
```

FINANCE VLAN 20

```
interface vlan 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
```

IT VLAN 30

```
interface vlan 30
 ip address 192.168.30.1 255.255.255.0
 no shutdown
exit
```

Management VLAN 40

```
interface vlan 40
 ip address 192.168.40.1 255.255.255.0
 no shutdown
exit
```

Server Room VLAN 50

```
interface vlan 50
 ip address 192.168.50.1 255.255.255.0
 no shutdown
exit
```

Attacker VLAN 99 (Compromised VLAN)

```
interface vlan 99
 ip address 192.168.99.1 255.255.255.0
 no shutdown
exit
```

```
CORE-SW>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES unset  down                  down
FastEthernet0/2        unassigned      YES unset  down                  down
FastEthernet0/3        unassigned      YES unset  down                  down
FastEthernet0/4        unassigned      YES unset  down                  down
FastEthernet0/5        unassigned      YES unset  down                  down
FastEthernet0/6        unassigned      YES unset  down                  down
FastEthernet0/7        unassigned      YES unset  down                  down
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down
FastEthernet0/22       unassigned      YES unset  down                  down
FastEthernet0/23       unassigned      YES unset  down                  down
FastEthernet0/24       unassigned      YES unset  down                  down
GigabitEthernet0/1     unassigned      YES unset  up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.1    YES manual up                    up
Vlan20                 192.168.20.1    YES manual up                    up
Vlan30                 192.168.30.1    YES manual up                    up
Vlan40                 192.168.40.1    YES manual up                    up
Vlan50                 192.168.50.1    YES manual up                    up
Vlan99                 192.168.99.1    YES manual up                    up
```

Trunk Configuration:

802.1Q encapsulation is configured BEFORE trunk mode.

For `Gi0/1`

```
interface gi0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
exit
```

For `Gi0/2`

```
interface gi0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
exit
```

```
CORE-SW>show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005
Gig0/2      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20,30,40,50,99
Gig0/2      1,10,20,30,40,50,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20,30,40,50,99
Gig0/2      1,10,20,30,40,50,99
```

Access Control Lists (ACLs)

HR ACL – VLAN 10

Purpose:

- Block HR ↔ Finance
- Allow other necessary traffic implicitly

```
ip access-list extended HR-ACL
deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
permit ip any any
```

Applied inbound:

```
interface vlan 10
ip access-group HR-ACL in
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Finance ACL – VLAN 20

Purpose:

- Block Finance ↔ HR
- Allow other necessary traffic implicitly

```
ip access-list extended FIN-ACL
deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
permit ip any any
```

Applied inbound:

```
interface vlan 20
ip access-group FIN-ACL in
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.40

Pinging 192.168.10.40 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.40:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.40.10: bytes=32 time=10ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms
```

Attacker ACL – VLAN 99

Purpose:

- Completely isolate the attacker's network

```
ip access-list extended BLOCK-ATTACKER
 deny ip 192.168.99.0 0.0.0.255 any
 deny ip any 192.168.99.0 0.0.0.255
 permit ip any any
```

Applied inbound:

```
interface vlan 99
ip access-group BLOCK-ATTACKER in
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.99.10

Pinging 192.168.99.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.99.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

SECURE REMOTE ACCESS TO CORE SWITCH

SSH Configuration on CORE-SW

The CORE-SW (Layer 3 switch) is hardened to allow secure remote administration using SSH Version 2. Telnet is disabled.

```
IP domain-name enterprise.lab
username ITADMIN privilege 15 secret Cisco@123
crypto key generate rsa modulus 1024
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
```

What This Does

- **ip domain-name** → Required for RSA key generation
- **username ITADMIN privilege 15** → Creates local admin account
- **crypto key generate rsa** → Enables SSH encryption
- **ip ssh version 2** → Forces secure SSHv2
- **Timeout & retries** → Reduces brute-force attempts

VTY Line Configuration

```
line vty 0 15
transport input ssh
login local
access-class SSH-MGMT in
```

What This Does

- **transport input ssh** → Disables Telnet completely
- **login local** → Uses local username database
- **access-class SSH-MGMT in** → Applies ACL inbound

SSH-Management ACL

```
ip access-list standard SSH-MGMT
```

```
permit 192.168.30.0 0.0.0.255
permit 192.168.40.0 0.0.0.255
deny any
```

This ensures **only IT and Management VLANs** can remotely manage the core switch.

```
CORE-SW>show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
```

Only these subnets can SSH to CORE-SW:

- VLAN 30 (IT Department)
- VLAN 40 (Management)

All other VLANs:

- HR ❌
- Finance ❌
- Servers ❌
- Attacker VLAN 99 ❌

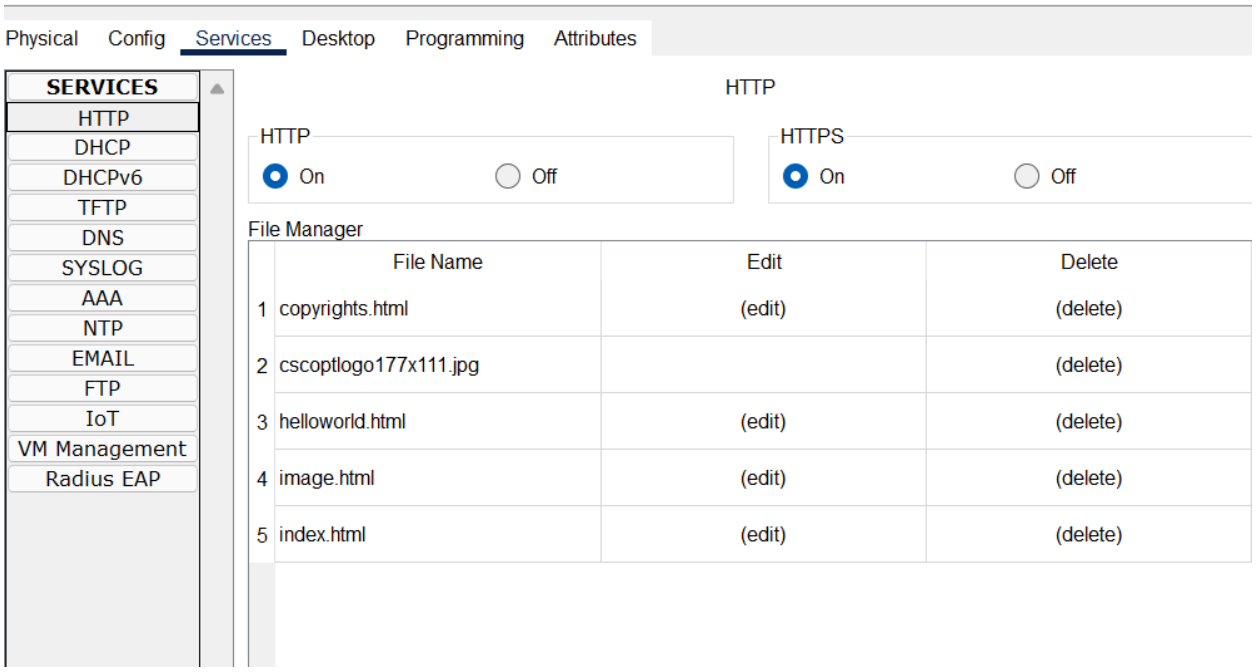
Security Validation

This configuration ensures:

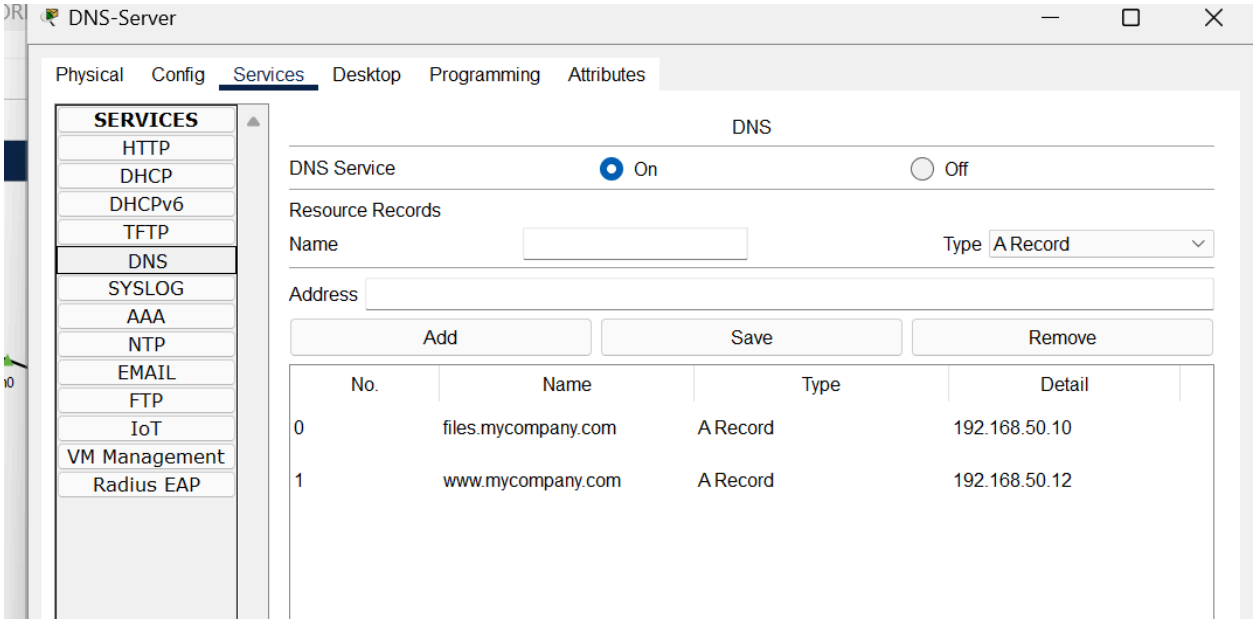
- SSH Version 2 is enforced
- Encrypted admin access
- Brute-force mitigation (timeouts & retries)
- No Telnet exposure

Endpoint & Server Configurations (Packet Tracer GUI)

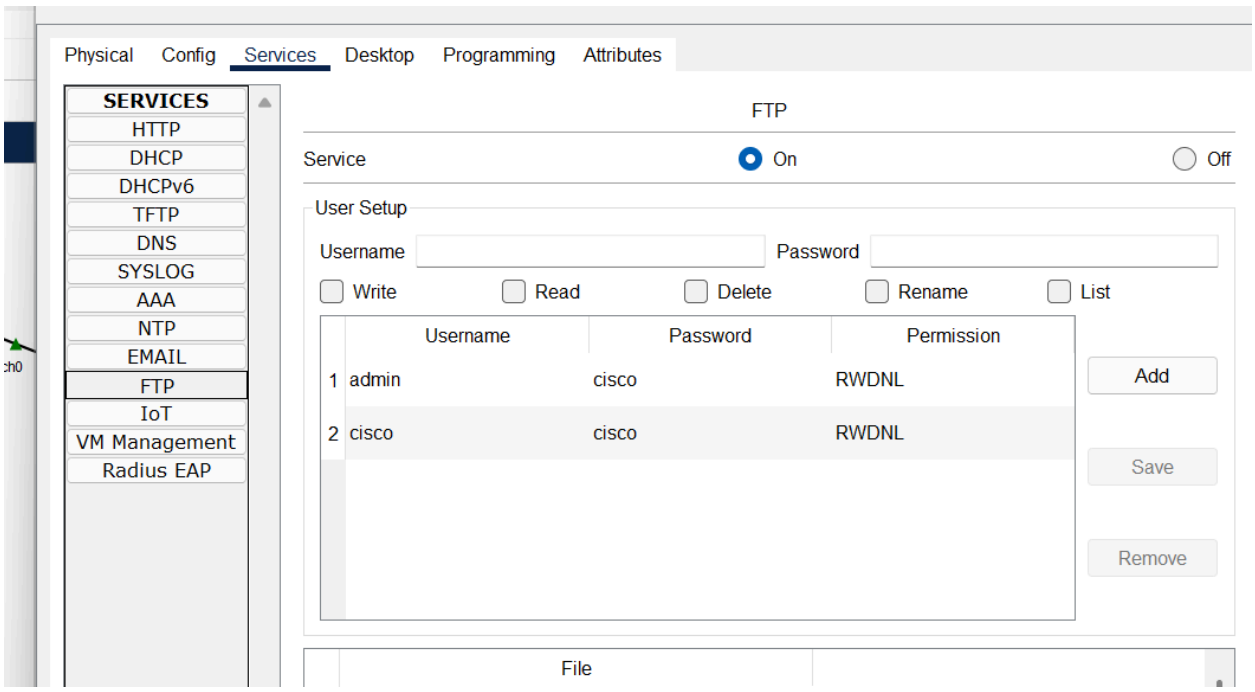
- **PCs/Laptops**: Desktop tab > IP Configuration > Static IP, Subnet, Gateway (per IP plan). DNS: 192.168.50.11.
- **Web Server (192.168.50.12)**: Services tab > HTTP (enable HTTP/HTTPS).



- **DNS Server (192.168.50.11)**: Services tab > DNS > Add A record (e.g., www.mycompany.com A 192.168.50.12).



- **File Server (192.168.50.10)**: Services tab > FTP (enable, add user for testing, e.g., ftpuser/password).



Conclusion

This project successfully demonstrates the design and implementation of a secure enterprise network architecture using VLAN segmentation, inter-VLAN routing, and Access Control List (ACL) enforcement within **Cisco Packet Tracer**.

Department-based VLAN segmentation effectively reduces broadcast domains and limits unauthorized communication between HR and Finance, preventing lateral movement across sensitive business units. Critical services (Web, DNS, and FTP) were isolated within a dedicated server VLAN, ensuring controlled access based on departmental roles.

Extended ACLs were strategically applied inbound on Switch Virtual Interfaces (SVIs) to enforce least-privilege access policies. The attacker VLAN (VLAN 99) was fully isolated at Layer 3, demonstrating containment of potentially compromised hosts and internal threat mitigation.

Secure remote management was implemented exclusively on the core multilayer switch using SSH Version 2, RSA encryption, authentication controls, and a management-specific ACL. This ensures that only authorized IT and Management subnets can administer the network infrastructure, eliminating insecure management protocols such as Telnet.

Inter-VLAN routing was performed on the **Cisco Catalyst 3560**, while endpoint connectivity was handled by **Cisco Catalyst 2960** access switches, reflecting a realistic hierarchical enterprise design model.

Through systematic validation testing (ping, web access, ACL verification, and SSH), all security objectives were met.

This simulation highlights core enterprise security principles:

- Network segmentation
- Least-privilege access control
- Management plane hardening
- Internal threat containment
- Proper ACL placement and verification

Overall, the project demonstrates practical understanding of enterprise network security design and provides a strong foundation for more advanced implementations involving firewalls, intrusion detection systems, and centralized logging platforms.